

Measure Theory – 1

Set Operations and De Morgan's Laws, Preimages of Functions,
Limits Superior and Inferior of Sets, and σ -Algebras

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Introduction

These notes cover the set-theoretic groundwork on which measure theory and rigorous probability are built: complements and set differences relative to a universal set Ω , De Morgan's laws for arbitrary families, preimages of functions and the operations they preserve, limits superior and inferior of sequences of sets, and σ -algebras together with the σ -algebras they generate.

1 Set Theory and Notation

1.1 Complements Relative to Ω

In measure theory and rigorous probability, a complement only makes sense relative to a fixed universal set (sample space) Ω .

- **Ambiguous:** A^c or A' when the underlying space is left implicit.
- **Rigorous:** $A^c = \Omega \setminus A = \{\omega \in \Omega : \omega \notin A\}$, with $A \subseteq \Omega$.

1.2 Set Differences

For $A, B \subseteq \Omega$,

$$A \setminus B = A \cap B^c = A \cap (\Omega \setminus B), \quad A \triangle B = (A \setminus B) \cup (B \setminus A). \quad (1)$$

Note that $A^c = \Omega \setminus A$ is the special case $B = A$, $A \rightarrow \Omega$ of the difference.

1.3 Indicator Functions

The **indicator** of $A \subseteq \Omega$ is $\mathbf{1}_A(\omega) = 1$ if $\omega \in A$ and 0 otherwise. Set operations translate into arithmetic:

$$\begin{aligned} \mathbf{1}_{A^c} &= 1 - \mathbf{1}_A, & \mathbf{1}_{A \cap B} &= \mathbf{1}_A \mathbf{1}_B, & \mathbf{1}_{A \cup B} &= \max(\mathbf{1}_A, \mathbf{1}_B), \\ \mathbf{1}_{A \setminus B} &= \mathbf{1}_A(1 - \mathbf{1}_B), & \mathbf{1}_{A \triangle B} &\equiv \mathbf{1}_A + \mathbf{1}_B \pmod{2}. \end{aligned}$$

Two sets are equal if and only if their indicators are equal, which often gives a quick proof of a set identity.

1.4 De Morgan's Laws and Distributivity

Proposition 1.1 (De Morgan's laws for arbitrary collections). *Let $\{A_i\}_{i \in I}$ be an indexed family of subsets of Ω (with I arbitrary, finite, countable, or uncountable). Then*

$$(1) \left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c,$$

$$(2) \left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c.$$

Proof. (1) We have

$$\begin{aligned} \omega \in \left(\bigcup_i A_i \right)^c &\iff \omega \notin A_i \text{ for every } i \in I \\ &\iff \omega \in A_i^c \text{ for every } i \in I \iff \omega \in \bigcap_i A_i^c. \end{aligned}$$

(2) Apply (1) to the family $\{A_i^c\}$ to get $\left(\bigcup_i A_i^c \right)^c = \bigcap_i A_i$, then take complements of both sides. $\square$

Proposition 1.2 (Distributive laws). *For $A \subseteq \Omega$ and a family $\{B_i\}_{i \in I}$,*

$$A \cap \bigcup_{i \in I} B_i = \bigcup_{i \in I} (A \cap B_i), \quad A \cup \bigcap_{i \in I} B_i = \bigcap_{i \in I} (A \cup B_i).$$

Proof. For the first identity, $\omega \in A \cap \bigcup_i B_i$ iff $\omega \in A$ and $\omega \in B_{i_0}$ for some i_0 , iff $\omega \in A \cap B_{i_0}$ for some i_0 . The second follows from the first and Proposition 1.1. $\square$

Example 1.3. Turning a union into a disjoint union is a standard technique. For any sequence $\{A_n\}$, define $D_1 = A_1$ and $D_n = A_n \setminus \bigcup_{k < n} A_k$. The D_n are pairwise disjoint, $D_n \subseteq A_n$, and $\bigcup_n D_n = \bigcup_n A_n$.

2 Functions, Preimages, and Inverse Images

Throughout, $f : \Omega \rightarrow \Omega'$ is a function between two spaces.

2.1 Definition of the Preimage

Definition 2.1 (Preimage / inverse image). For $B \subseteq \Omega'$,

$$f^{-1}(B) = \{\omega \in \Omega : f(\omega) \in B\}. \quad (2)$$

For a collection $\mathcal{H}$ of subsets of Ω' we write $f^{-1}(\mathcal{H}) = \{f^{-1}(B) : B \in \mathcal{H}\}$.

Remark. The map $f^{-1} : \mathcal{P}(\Omega') \rightarrow \mathcal{P}(\Omega)$ is always well defined as an operation on sets, even when f is neither injective nor surjective, and even though f need not have an inverse as a point map.

2.2 Preservation Properties

Theorem 2.2 (Preimages preserve set operations). *Let $B, B_1, B_2 \subseteq \Omega'$ and $\{B_i\}_{i \in I}$ be any family of subsets of Ω' . Then*

- (a) $f^{-1}(\Omega') = \Omega$ and $f^{-1}(\emptyset) = \emptyset$;
- (b) $f^{-1}(B^c) = (f^{-1}(B))^c$;
- (c) $f^{-1}(\bigcup_i B_i) = \bigcup_i f^{-1}(B_i)$;
- (d) $f^{-1}(\bigcap_i B_i) = \bigcap_i f^{-1}(B_i)$;
- (e) $f^{-1}(B_1 \setminus B_2) = f^{-1}(B_1) \setminus f^{-1}(B_2)$.

Proof. (a) Every ω satisfies $f(\omega) \in \Omega'$, and none satisfies $f(\omega) \in \emptyset$.

(b) *Complements.* For $B \subseteq \Omega'$,

$$\begin{aligned} \omega \in f^{-1}(B^c) &\iff f(\omega) \in B^c \\ &\iff f(\omega) \notin B \\ &\iff \omega \notin f^{-1}(B) \\ &\iff \omega \in (f^{-1}(B))^c. \end{aligned}$$

(c) *Arbitrary unions.*

$$\begin{aligned} \omega \in f^{-1}\left(\bigcup_{i \in I} B_i\right) &\iff f(\omega) \in \bigcup_{i \in I} B_i \\ &\iff \exists i_0 \in I : f(\omega) \in B_{i_0} \\ &\iff \exists i_0 \in I : \omega \in f^{-1}(B_{i_0}) \\ &\iff \omega \in \bigcup_{i \in I} f^{-1}(B_i). \end{aligned}$$

(d) *Arbitrary intersections.*

$$\begin{aligned} \omega \in f^{-1}\left(\bigcap_{i \in I} B_i\right) &\iff f(\omega) \in B_i \quad \forall i \in I \\ &\iff \omega \in f^{-1}(B_i) \quad \forall i \in I \iff \omega \in \bigcap_{i \in I} f^{-1}(B_i). \end{aligned}$$

(e) *Set differences.* Using $B_1 \setminus B_2 = B_1 \cap B_2^c$ together with (d) and (b),

$$f^{-1}(B_1 \setminus B_2) = f^{-1}(B_1) \cap f^{-1}(B_2^c) = f^{-1}(B_1) \cap (f^{-1}(B_2))^c = f^{-1}(B_1) \setminus f^{-1}(B_2). \quad \square$$

Corollary 2.3 (Further properties). (i) **Monotone:** $B_1 \subseteq B_2 \implies f^{-1}(B_1) \subseteq f^{-1}(B_2)$.

(ii) **Disjointness:** if $B_1 \cap B_2 = \emptyset$, then $f^{-1}(B_1) \cap f^{-1}(B_2) = \emptyset$. In particular, preimages of a partition of Ω' form a partition of Ω (after discarding empty sets).

(iii) **Composition:** for $g : \Omega' \rightarrow \Omega''$, $(g \circ f)^{-1}(C) = f^{-1}(g^{-1}(C))$.

Proof. (i) follows from (d): $B_1 \subseteq B_2$ means $B_1 = B_1 \cap B_2$. (ii) is (d) with (a). (iii): $\omega \in (g \circ f)^{-1}(C) \iff g(f(\omega)) \in C \iff f(\omega) \in g^{-1}(C) \iff \omega \in f^{-1}(g^{-1}(C))$. $\square$

2.3 Images Do Not Behave as Nicely

Preimages preserve *all* the operations above, but forward images $f(A) = \{f(\omega) : \omega \in A\}$ preserve only unions.

Example 2.4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$.

- **Intersections:** $A = [-2, -1]$, $B = [1, 2]$ have $A \cap B = \emptyset$, yet $f(A) \cap f(B) = [1, 4] \neq \emptyset = f(A \cap B)$.
- **Complements:** $A = [0, \infty)$ gives $f(A^c) = (0, \infty)$ but $f(A)^c = (-\infty, 0)$.

This is exactly why measurability is phrased through preimages.

2.4 Preimages of σ -Algebras

Proposition 2.5. *If $\mathcal{G}$ is a σ -algebra on Ω' , then $f^{-1}(\mathcal{G}) = \{f^{-1}(B) : B \in \mathcal{G}\}$ is a σ -algebra on Ω .*

Proof. $\Omega = f^{-1}(\Omega') \in f^{-1}(\mathcal{G})$. If $f^{-1}(B) \in f^{-1}(\mathcal{G})$, then $(f^{-1}(B))^c = f^{-1}(B^c)$ with $B^c \in \mathcal{G}$. If $B_n \in \mathcal{G}$, then $\bigcup_n f^{-1}(B_n) = f^{-1}(\bigcup_n B_n)$ with $\bigcup_n B_n \in \mathcal{G}$. All three axioms follow from Theorem 2.2. $\square$

Remark. A function $f : (\Omega, \mathcal{F}) \rightarrow (\Omega', \mathcal{G})$ is called **measurable** if $f^{-1}(\mathcal{G}) \subseteq \mathcal{F}$. Proposition 2.5 says $f^{-1}(\mathcal{G})$ is the smallest σ -algebra on Ω making f measurable.

3 Sequences of Sets: Limits Superior and Inferior

Let $\{A_n\}_{n=1}^\infty$ be a sequence of subsets of Ω .

3.1 Definitions

Definition 3.1 (Limit superior). The limit superior collects the elements belonging to infinitely many A_n (“infinitely often”):

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^\infty \bigcup_{k=n}^\infty A_k = \{A_n \text{ i.o.}\}. \quad (3)$$

Definition 3.2 (Limit inferior). The limit inferior collects the elements belonging to all but finitely many A_n (“eventually”):

$$\liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^\infty \bigcap_{k=n}^\infty A_k = \{A_n \text{ ev.}\}. \quad (4)$$

Pointwise meaning.

- $\omega \in \limsup A_n \iff \forall n \exists k \geq n : \omega \in A_k \iff \omega \in A_k \text{ for infinitely many } k$.
- $\omega \in \liminf A_n \iff \exists n \forall k \geq n : \omega \in A_k \iff \omega \notin A_k \text{ for only finitely many } k$.

3.2 Basic Properties

Proposition 3.3. *For any sequence $\{A_n\}$:*

- (a) $\liminf_{n \rightarrow \infty} A_n \subseteq \limsup_{n \rightarrow \infty} A_n$;
- (b) **Duality:** $(\limsup A_n)^c = \liminf A_n^c$ and $(\liminf A_n)^c = \limsup A_n^c$;
- (c) **Indicators:** $\mathbf{1}_{\limsup A_n} = \limsup_n \mathbf{1}_{A_n}$ and $\mathbf{1}_{\liminf A_n} = \liminf_n \mathbf{1}_{A_n}$.

Proof. (a) If ω lies in all but finitely many A_n , it certainly lies in infinitely many (there are infinitely many indices). Formally: choose n_0 with $\omega \in \bigcap_{k \geq n_0} A_k$; then for every n , $\omega \in A_{\max(n, n_0)} \subseteq \bigcup_{k \geq n} A_k$.

(b) By Proposition 1.1,

$$\left(\bigcap_n \bigcup_{k \geq n} A_k \right)^c = \bigcup_n \left(\bigcup_{k \geq n} A_k \right)^c = \bigcup_n \bigcap_{k \geq n} A_k^c = \liminf A_n^c.$$

The second identity is the same computation with the roles exchanged.

(c) $\omega \in A_k$ infinitely often iff $\mathbf{1}_{A_k}(\omega) = 1$ for infinitely many k iff $\limsup_k \mathbf{1}_{A_k}(\omega) = 1$; similarly for $\liminf$. $\square$

Proposition 3.4 (Compatibility with unions and intersections).

$$\limsup_n (A_n \cup B_n) = \limsup_n A_n \cup \limsup_n B_n, \quad \liminf_n (A_n \cap B_n) = \liminf_n A_n \cap \liminf_n B_n.$$

Proof. If ω is in infinitely many $A_k \cup B_k$, then it is in infinitely many A_k or infinitely many B_k (an infinite set of indices split into two parts has an infinite part), and the converse is clear. For $\liminf$: if ω lies in all A_k for $k \geq n_1$ and all B_k for $k \geq n_2$, then it lies in all $A_k \cap B_k$ for $k \geq \max(n_1, n_2)$; the reverse inclusion is immediate. $\square$

Definition 3.5 (Set-theoretic limit). If $\liminf A_n = \limsup A_n$, this common set is called the limit of the sequence and denoted $\lim_n A_n$.

Example 3.6. (1) **Monotone sequences.** If $A_n \uparrow$ (i.e. $A_n \subseteq A_{n+1}$), then $\lim A_n = \bigcup_n A_n$. If $A_n \downarrow$, then $\lim A_n = \bigcap_n A_n$.

(2) **Alternating intervals.** Let $A_n = [0, 1]$ for n even and $A_n = [1, 2]$ for n odd. Then $\limsup A_n = [0, 2]$ (every point of $[0, 2]$ is in infinitely many A_n), while $\liminf A_n = [0, 1] \cap [1, 2] = \{1\}$. The inclusion in Proposition 3.3(a) is strict, and the limit does not exist.

(3) **Shrinking intervals.** $A_n = (-\frac{1}{n}, \frac{1}{n})$ is decreasing, so $\lim A_n = \bigcap_n A_n = \{0\}$.

4 σ -Algebras and Generated Systems

4.1 Definition of a σ -Algebra

Definition 4.1 (σ -algebra). A collection $\mathcal{F}$ of subsets of Ω is a σ -algebra on Ω if

- (i) $\Omega \in \mathcal{F}$;
- (ii) $A \in \mathcal{F} \implies A^c = \Omega \setminus A \in \mathcal{F}$ (closure under complementation);

(iii) $A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ (closure under countable unions).

The pair $(\Omega, \mathcal{F})$ is a **measurable space**.

Proposition 4.2 (Immediate consequences). *If $\mathcal{F}$ is a σ -algebra on Ω , then*

- (a) $\emptyset \in \mathcal{F}$;
- (b) $\mathcal{F}$ is closed under finite unions;
- (c) $\mathcal{F}$ is closed under finite and countable intersections;
- (d) $A, B \in \mathcal{F} \implies A \setminus B \in \mathcal{F}$ and $A \triangle B \in \mathcal{F}$;
- (e) $\limsup A_n$ and $\liminf A_n$ are in $\mathcal{F}$ whenever each $A_n \in \mathcal{F}$.

Proof. (a) $\emptyset = \Omega^c$. (b) Put $A_i = \emptyset$ for $i > m$ in (iii). (c) $\bigcap_i A_i = \left(\bigcup_i A_i^c\right)^c$ by De Morgan. (d) $A \setminus B = A \cap B^c$, and $A \triangle B = (A \setminus B) \cup (B \setminus A)$. (e) Both are built from A_n by countable unions and intersections. $\square$

Example 4.3. • **Trivial:** $\{\emptyset, \Omega\}$, the smallest σ -algebra.

- **Power set:** $\mathcal{P}(\Omega)$, the largest.
- **One generator:** $\sigma(\{A\}) = \{\emptyset, A, A^c, \Omega\}$.
- **Countable/co-countable:** on an uncountable Ω , the sets that are countable or have countable complement.
- **Non-example (algebra but not σ -algebra):** finite and co-finite subsets of $\mathbb{N}$. It is closed under finite unions and complements, but the even numbers are a countable union of singletons and are neither finite nor co-finite.
- **Non-example (union of σ -algebras):** on $\Omega = \{1, 2, 3\}$, $\mathcal{F}_1 = \sigma(\{1\})$ and $\mathcal{F}_2 = \sigma(\{2\})$ each are σ -algebras, but $\mathcal{F}_1 \cup \mathcal{F}_2$ contains $\{1\}$ and $\{2\}$ and not $\{1, 2\}$.

Proposition 4.4 (Intersections of σ -algebras). *If $\{\mathcal{F}_\alpha\}_{\alpha \in A}$ is any (non-empty) family of σ -algebras on Ω , then $\bigcap_\alpha \mathcal{F}_\alpha$ is a σ -algebra.*

Proof. $\Omega \in \mathcal{F}_\alpha$ for each α . If A is in every $\mathcal{F}_\alpha$, so is A^c . If $A_1, A_2, \dots$ are in every $\mathcal{F}_\alpha$, so is their union. $\square$

4.2 Generated σ -Algebras

Definition 4.5 (Generated σ -algebra). For $\mathcal{C} \subseteq \mathcal{P}(\Omega)$,

$$\sigma(\mathcal{C}) = \bigcap \{ \mathcal{G} \subseteq \mathcal{P}(\Omega) : \mathcal{G} \text{ is a } \sigma\text{-algebra and } \mathcal{C} \subseteq \mathcal{G} \}.$$

The family is non-empty (it contains $\mathcal{P}(\Omega)$), so by Proposition 4.4 $\sigma(\mathcal{C})$ is a σ -algebra. It is the **smallest** σ -algebra containing $\mathcal{C}$.

Verifying that a candidate $\mathcal{H}$ equals $\sigma(\mathcal{C})$. Prove a double inclusion:

1. $\mathcal{H} \subseteq \sigma(\mathcal{C})$: show every element of $\mathcal{H}$ must belong to *every* σ -algebra containing $\mathcal{C}$.
2. $\sigma(\mathcal{C}) \subseteq \mathcal{H}$: show that $\mathcal{H}$ is itself a σ -algebra containing $\mathcal{C}$ (then minimality gives the inclusion).

Example 4.6 (Finite partition generator). Let $\mathcal{P} = \{E_1, \dots, E_k\}$ be a partition of Ω . Then

$$\sigma(\mathcal{P}) = \left\{ \bigcup_{i \in I} E_i : I \subseteq \{1, \dots, k\} \right\} =: \mathcal{H}.$$

Proof. ($\mathcal{H} \subseteq \sigma(\mathcal{P})$) Any σ -algebra containing the E_i contains all their finite unions. ($\sigma(\mathcal{P}) \subseteq \mathcal{H}$) $\mathcal{H}$ contains each E_i and Ω (take $I = \{1, \dots, k\}$). The complement of $\bigcup_{i \in I} E_i$ is $\bigcup_{i \notin I} E_i \in \mathcal{H}$, using disjointness and $\bigcup E_i = \Omega$. A countable union of members of $\mathcal{H}$ is a union of E_i over $\bigcup_n I_n \subseteq \{1, \dots, k\}$, so it lies in $\mathcal{H}$. Hence $\mathcal{H}$ is a σ -algebra containing $\mathcal{P}$, so $\sigma(\mathcal{P}) \subseteq \mathcal{H}$. $\square$

In particular $|\sigma(\mathcal{P})| = 2^k$, and the E_i are the **atoms**: no non-empty proper subset of E_i is in $\sigma(\mathcal{P})$.

Example 4.7 (Two generating sets). $\sigma(\{A, B\})$ is generated by the four-set partition $A \cap B$, $A \cap B^c$, $A^c \cap B$, $A^c \cap B^c$ (some possibly empty), so it has at most $2^4 = 16$ members.

Example 4.8 (Borel σ -algebra). $\mathcal{B}(\mathbb{R}) = \sigma(\{(a, b] : a < b\})$. It equals the σ -algebra generated by the open sets, by the closed sets, by the open intervals, or by the half-lines $(-\infty, x]$. For instance, $(a, b) = \bigcup_n (a, b - \frac{1}{n}]$ and $\{x\} = \bigcap_n (x - \frac{1}{n}, x]$ show that open intervals and points are Borel sets, so every countable set is Borel.

Some Key Learnings:

- Closure under finite unions alone gives an *algebra*, not a σ -algebra.
- A union of σ -algebras is generally not a σ -algebra; an intersection always is.
- $\sigma(\mathcal{C})$ is usually much larger than $\mathcal{C}$ (and than the closure of $\mathcal{C}$ under a single round of complements and unions); always argue via minimality.